

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): DHD800

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.04.0000-b007	2.01.0001-b001	1.2.1

Version History

Module Version	Date	Notes
1_0_2_0	1/3/2017	Fixed authentication issues
1_0_0_0	9/8/2016	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example:
InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example:
InterfaceName.connectionCounter = 5

Control Commands, States, and Qualifiers:

AspectRatio	'Normal' 'True'	'Full' 'Custom'	'Zoom' 'Natural'
AutoImage	None		
ClosedCaption	'CC1' 'CC4'	'CC2' 'Off'	'CC3'
ExecutiveMode	'Mode 1'	'Mode 2'	'Off'
Focus	'In'	'Out'	
Freeze	'On'	'Off'	
Input	'Input 3 RGB' 'Input 1 DVI (HDCP)' 'Input 2 Video' 'Input 2 YPbPr' 'Input 3 YCbCr' 'Input 3 SDI-2' 'Input 3 SCART' 'Input 4 RGB' 'Input 4 YCbCr' 'Input 4 SDI-2' 'Input 4 HDMI'	'Input 1 Ana. RGB' 'Input 1 SCART' 'Input 2 S-Video' 'Input 3 Video' 'Input 3 YPbPr' 'Input 3 Digital' 'Input 3 HDMI' 'Input 4 Video' 'Input 4 YPbPr' 'Input 4 HDCP'	'Input 1 DVI (Digital)' 'Input 1 HDMI' 'Input 2 YCbCr' 'Input 3 S-Video' 'Input 3 SDI-1' 'Input 3 HDCP' 'Input 4 DVI-D' 'Input 4 S-Video' 'Input 4 SDI-1' 'Input 4 SCART'
LampMode	'Full'	'Lamp 1'	'Lamp 2'
LampPower	'Normal'	'Eco 1'	'Eco 2'
LensShift	'Up' 'Right'	'Down'	'Left'
MenuNavigation	'Up' 'Right' 'Menu Off'	'Down' 'Enter'	'Left' 'Menu On'
Power	'On'	'Off'	
VideoMute	'On'	'Off'	
Zoom	'In'	'Out'	

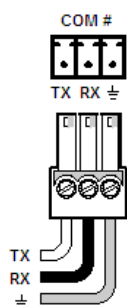
Status Available:

AspectRatio	'Normal' 'True'	'Full' 'Custom'	'Zoom' 'Natural'
ClosedCaption	'CC1' 'CC4'	'CC2' 'Off'	'CC3'
ConnectionStatus	'Connected' 'Disconnected'		
DeviceStatus	'Normal' 'Cooling Down (Lamp Failure)' 'Invalid Command' 'Standby after Cooling Down (Shutter Mgmt)'	'Power Fail' 'Cooling Down (Abnormal Temp)' 'Process Cooling Down after Off (Shutter Mgmt)' 'Power Save'	'Power Save (Cooling Down)' 'Standby after Cooling Down (Abnormal Temp)' 'Standby after Cooling Down (Lamp Failure)'
ExecutiveMode	'Mode 1'	'Mode 2'	'Off'
FilterUsage	Hours		
Freeze	'On'	'Off'	
Input	'Input 3 RGB' 'Input 1 DVI (HDCP)' 'Input 2 Video' 'Input 2 YPbPr' 'Input 3 YCbCr' 'Input 3 SDI-2' 'Input 3 SCART' 'Input 4 RGB' 'Input 4 YCbCr' 'Input 4 SDI-2' 'Input 4 HDMI'	'Input 1 Ana. RGB' 'Input 1 SCART' 'Input 2 S-Video' 'Input 3 Video' 'Input 3 YPbPr' 'Input 3 Digital' 'Input 3 HDMI' 'Input 4 Video' 'Input 4 YPbPr' 'Input 4 HDCP'	'Input 1 DVI (Digital)' 'Input 1 HDMI' 'Input 2 YCbCr' 'Input 3 S-Video' 'Input 3 SDI-1' 'Input 3 HDCP' 'Input 4 DVI-D' 'Input 4 S-Video' 'Input 4 SDI-1' 'Input 4 SCART'
LampMode	'Full'	'Lamp 1'	'Lamp 2'
LampPower	'Normal'	'Eco 1'	'Eco 2'
LampStatus	'Lamp On'	'Lamp Off'	'Lamp Failure'
['LampSelect']	'1' – '2'		
LampUsage	Hours		
['LampSelect']	'1' – '2'		
OperationHours	0 – 99999		
Power	'On' 'Cooling Down'	'Off'	'Warming Up'

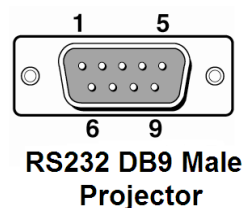
VideoMute	'On'	'Off'
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Cable and Adapter Requirements:

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device:**Serial communication:****Port Type:** RS-232**Baud Rate:** 19200**Data Bits:** 8**Parity:** None**Stop Bits:** One**Flow Control:** None**Pin Assignments Diagram**

Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND

**General Notes:**

Network communication:

When configuring the Ethernet module, be sure device settings match that of the GCP configuration.

Port Type:	Ethernet
Default Port:	10000
Logon Credentials Supported:	Yes
Default Password:	0000
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Appendix A. Set Commands

Aspect Ratio Custom	CF SCREEN CUSTOM\x0D
Aspect Ratio Full	CF SCREEN Full\x0D
Aspect Ratio Natural	CF SCREEN NATURAL\x0D
Aspect Ratio Normal	CF SCREEN NORMAL\x0D
Aspect Ratio True	CF SCREEN TRUE\x0D
Aspect Ratio Zoom	CF SCREEN ZOOM\x0D
Auto Image None	C89\x0D
Closed Caption CC1	CF CCAPTIONDISP CC1\x0D
Closed Caption CC2	CF CCAPTIONDISP CC2\x0D
Closed Caption CC3	CF CCAPTIONDISP CC3\x0D
Closed Caption CC4	CF CCAPTIONDISP CC4\x0D
Closed Caption Off	CF CCAPTIONDISP OFF\x0D
Executive Mode Mode 1	CF KEYDIS KEY\x0D
Executive Mode Mode 2	CF KEYDIS RC\x0D
Executive Mode Off	CF KEYDIS NONE\x0D
Focus In	C4A\x0D
Focus Out	C4B\x0D
Freeze Off	CF FREEZE OFF\x0D
Freeze On	CF FREEZE ON\x0D
Input Input 1 Ana. RGB	CF INPUT1 ANALOG\x0D
Input Input 1 DVI (Digital)	CF INPUT1 DIGITAL\x0D
Input Input 1 DVI (HDCP)	CF INPUT1 HDCP\x0D
Input Input 1 HDMI	CF INPUT1 HDMI\x0D
Input Input 1 SCART	CF INPUT1 SCART\x0D
Input Input 2 S-Video	CF INPUT2 S-VIDEO\x0D
Input Input 2 Video	CF INPUT2 VIDEO\x0D
Input Input 2 YCbCr	CF INPUT2 YCBCR\x0D
Input Input 2 YPbPr	CF INPUT2 YPBPR\x0D
Input Input 3 Digital	CF INPUT3 DIGITAL\x0D
Input Input 3 HDCP	CF INPUT3 HDCP\x0D
Input Input 3 HDMI	CF INPUT3 HDMI\x0D
Input Input 3 RGB	CF INPUT3 ANALOG\x0D
Input Input 3 SCART	CF INPUT3 SCART\x0D
Input Input 3 SDI-1	CF INPUT3 S-SDI1\x0D
Input Input 3 SDI-2	CF INPUT3 S-SDI2\x0D
Input Input 3 S-Video	CF INPUT3 S-VIDEO\x0D
Input Input 3 Video	CF INPUT3 VIDEO\x0D
Input Input 3 YCbCr	CF INPUT3 YCBCR\x0D
Input Input 3 YPbPr	CF INPUT3 YPBPR\x0D
Input Input 4 DVI-D	CF INPUT4 DIGITAL\x0D
Input Input 4 HDCP	CF INPUT4 HDCP\x0D
Input Input 4 HDMI	CF INPUT4 HDMI\x0D
Input Input 4 RGB	CF INPUT4 ANALOG\x0D
Input Input 4 SCART	CF INPUT4 SCART\x0D
Input Input 4 SDI-1	CF INPUT4 SDI1\x0D
Input Input 4 SDI-2	CF INPUT4 SDI2\x0D
Input Input 4 S-Video	CF INPUT4 S-VIDEO\x0D

Input Input 4 Video	CF INPUT4 VIDEO\x0D
Input Input 4 YCbCr	CF INPUT4 YCBCR\x0D
Input Input 4 YPbPr	CF INPUT4 YPBPR\x0D
Lamp Mode Full	CF LAMPMODE FULL\x0D
Lamp Mode Lamp 1	CF LAMPMODE LAMP1\x0D
Lamp Mode Lamp 2	CF LAMPMODE LAMP2\x0D
Lamp Power Eco 1	CF AUTOLAMPCTRL ECO1\x0D
Lamp Power Eco 2	CF AUTOLAMPCTRL ECO2\x0D
Lamp Power Normal	CF AUTOLAMPCTRL NORMAL\x0D
Lens Shift Down	C5E\x0D
Lens Shift Left	C5F\x0D
Lens Shift Right	C60\x0D
Lens Shift Up	C5D\x0D
Menu Navigation Down	CF KEYEMU DN\x0D
Menu Navigation Enter	CF KEYEMU SELECT\x0D
Menu Navigation Left	CF KEYEMU LEFT\x0D
Menu Navigation Menu Off	CF MENU OFF\x0D
Menu Navigation Menu On	CF MENU ON\x0D
Menu Navigation Right	CF KEYEMU RIGHT\x0D
Menu Navigation Up	CF KEYEMU UP\x0D
Power Off	CF POWER OFF\x0D
Power On	CF POWER ON\x0D
Video Mute Off	CF VMUTE OFF\x0D
Video Mute On	CF VMUTE ON\x0D
Zoom In	C47\x0D
Zoom Out	C46\x0D

Appendix B. Query Commands

Aspect Ratio	CR SCREEN\x0D
Closed Caption	CR CCAPTIONDISP\x0D
Executive Mode	CR KEYDIS\x0D
Filter Usage	CR FILH\x0D
Freeze	CR FREEZE\x0D
Input	CR SOURCE\x0D
Lamp Mode	CR LAMPMODE\x0D
Lamp Power	CR AUTOLAMPCTRL\x0D
Lamp Status LampSelect 1	CR LAMPSTS\x0D
Lamp Status LampSelect 2	CR LAMPSTS\x0D
Lamp Usage LampSelect 1	CR LAMPH\x0D
Lamp Usage LampSelect 2	CR LAMPH\x0D
Operation Hours	CR PROJH\x0D
Power	CR STATUS\x0D
Video Mute	CR VMUTE\x0D